Formally Verified

HRT challenge: Simulated Market Price Prediction Challenge

Data Science Rule No. 1

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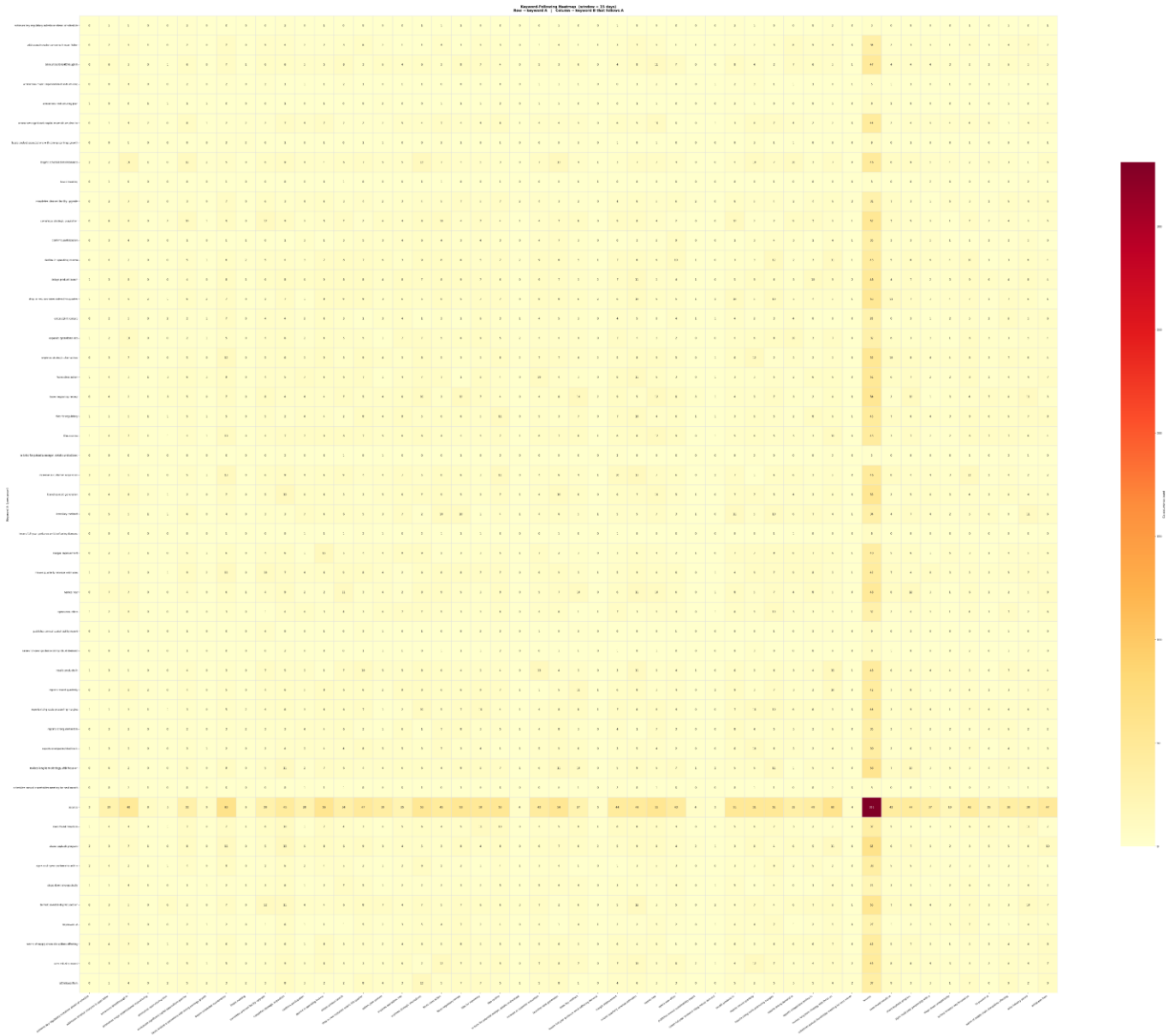
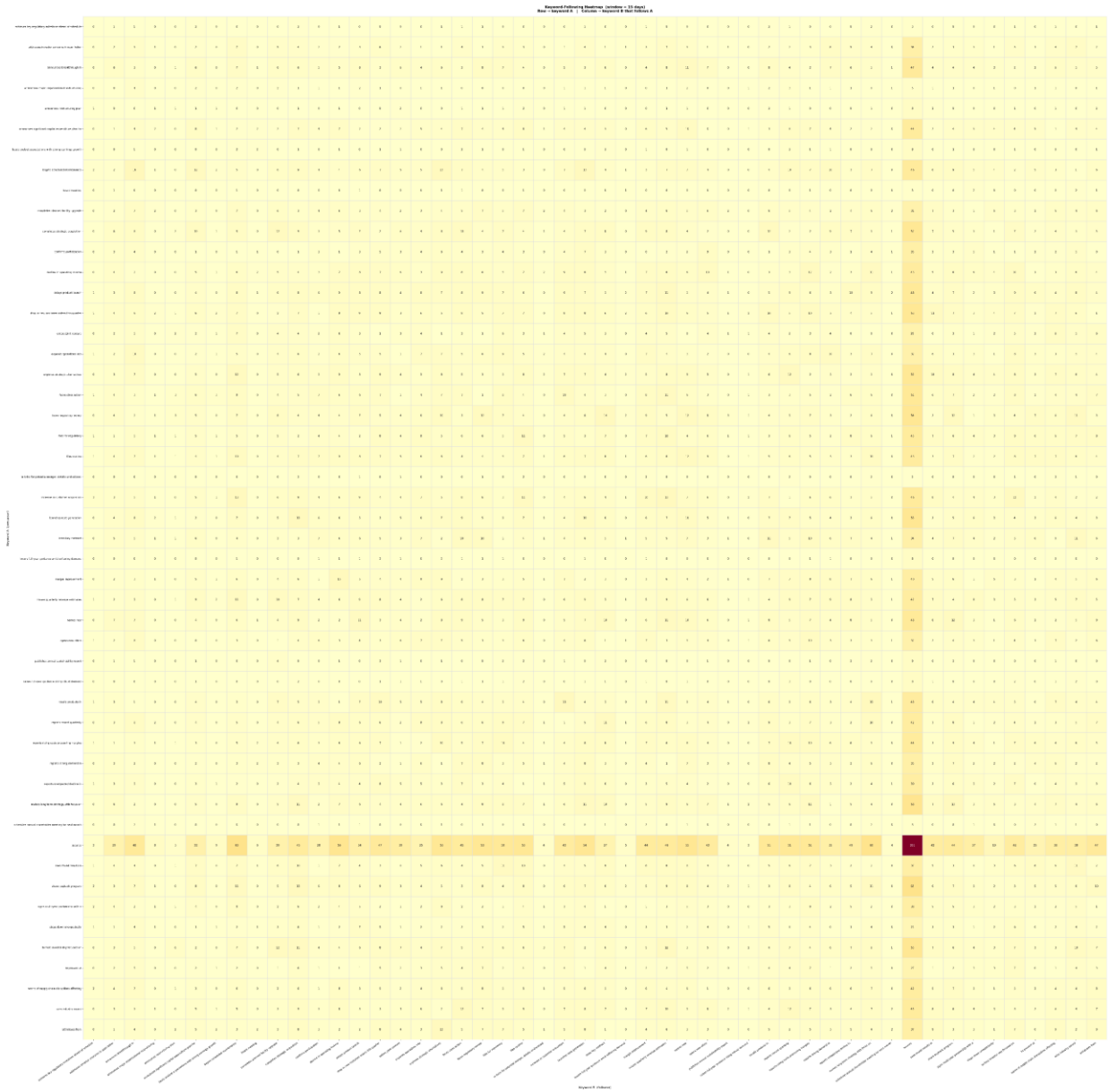
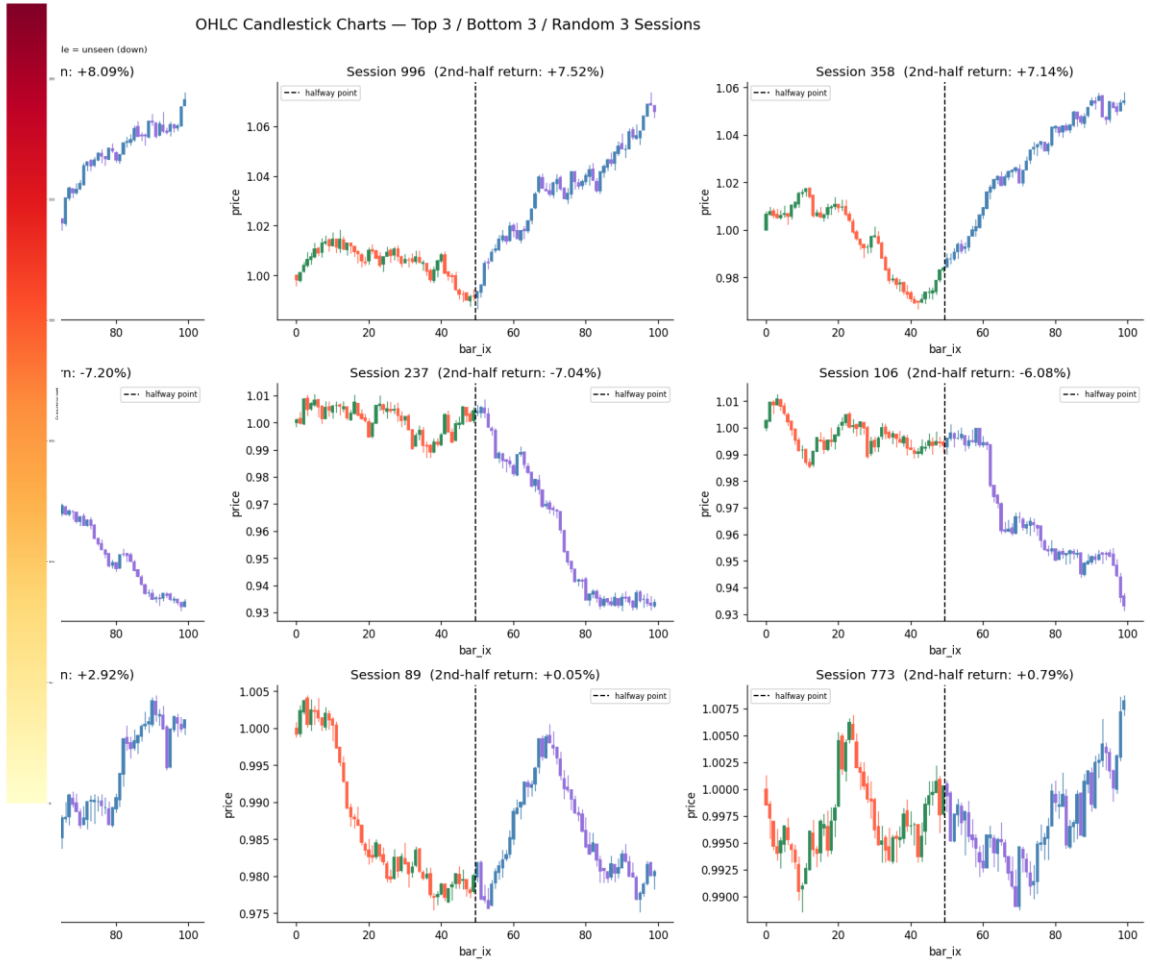
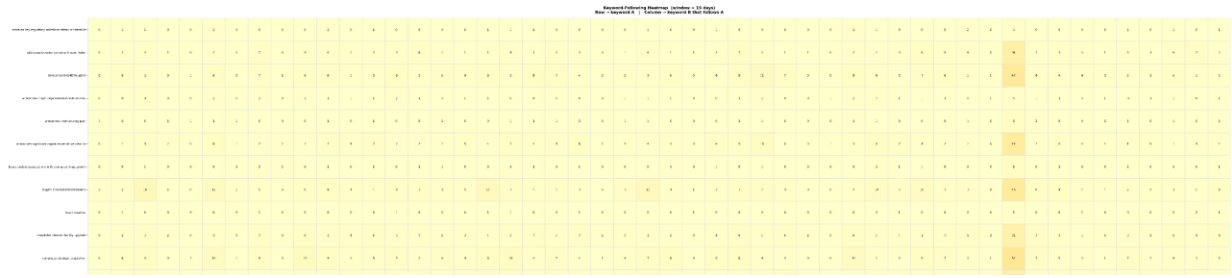
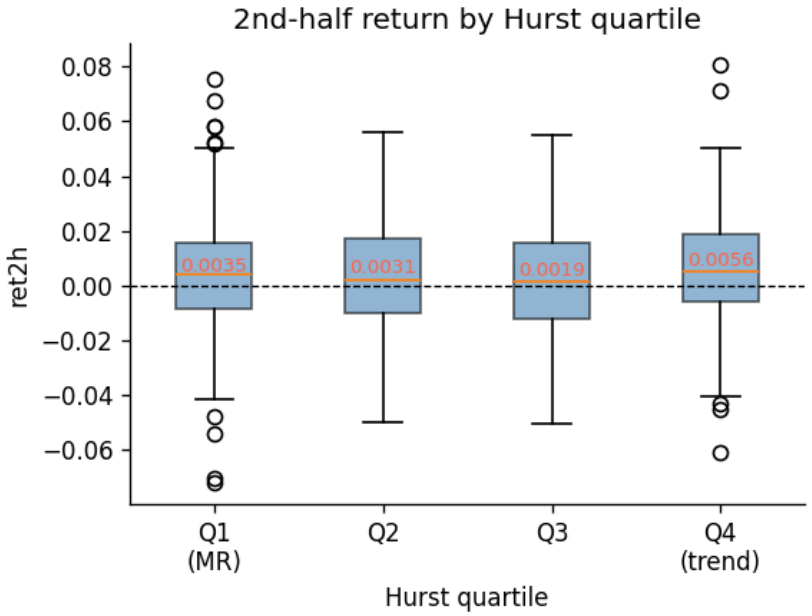

Keyword Following Heatmap (window = 15 days)
Row = keyword A | Column = keyword B that follows A
Keyword A (Source)
Keyword B (Target)
Count within window

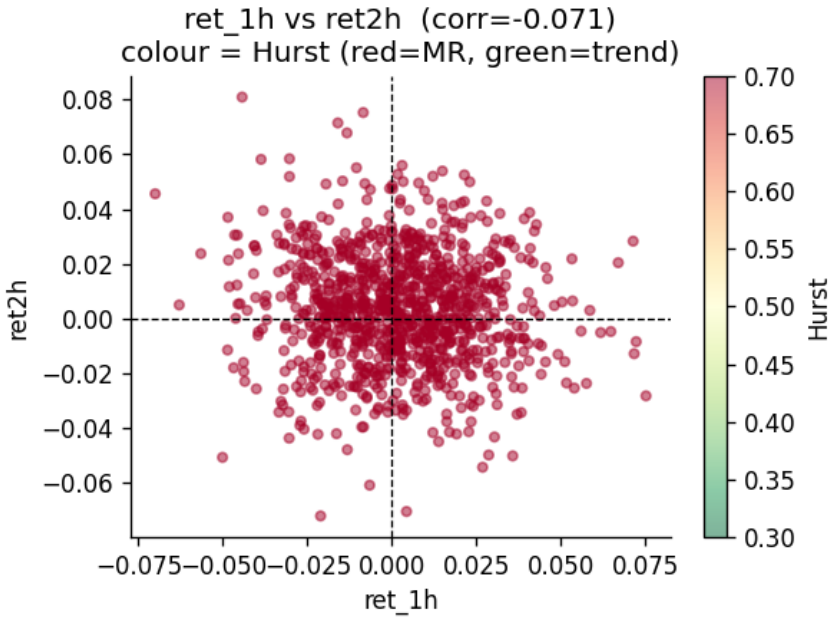


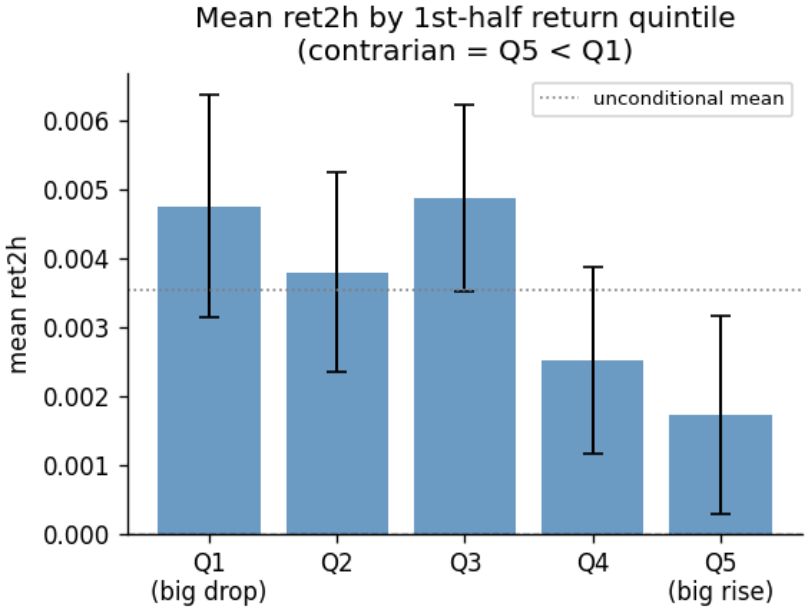

OHLC Candlestick Charts — Top 3 / Bottom 3 / Random 3 Sessions
le = unseen (down)
n: +8.09%)
Session 996 (2nd-half return: +7.52%)
Session 358 (2nd-half return: +7.14%)
n: -7.20%)
Session 237 (2nd-half return: -7.04%)
Session 106 (2nd-half return: -6.08%)
n: +2.92%)
Session 89 (2nd-half return: +0.05%)
Session 773 (2nd-half return: +0.79%)
halfway point
price
bar_ix

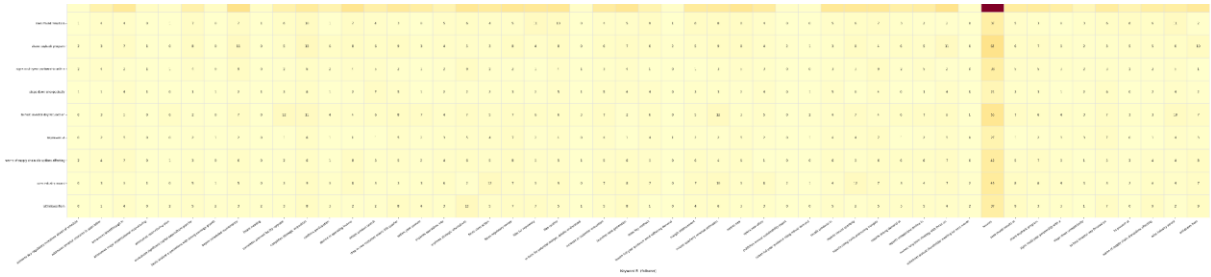


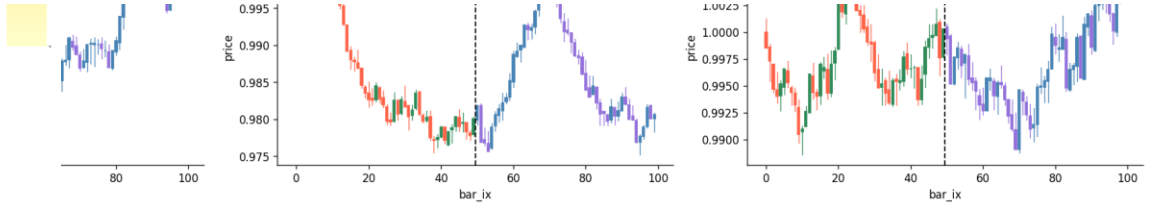
OHLC Candlestick Charts — Top 3 / Bottom 3 / Random 3 Sessions
le = unseen (down)
n: +8.09%)
Session 996 (2nd-half return: +7.52%)
Session 358 (2nd-half return: +7.14%)

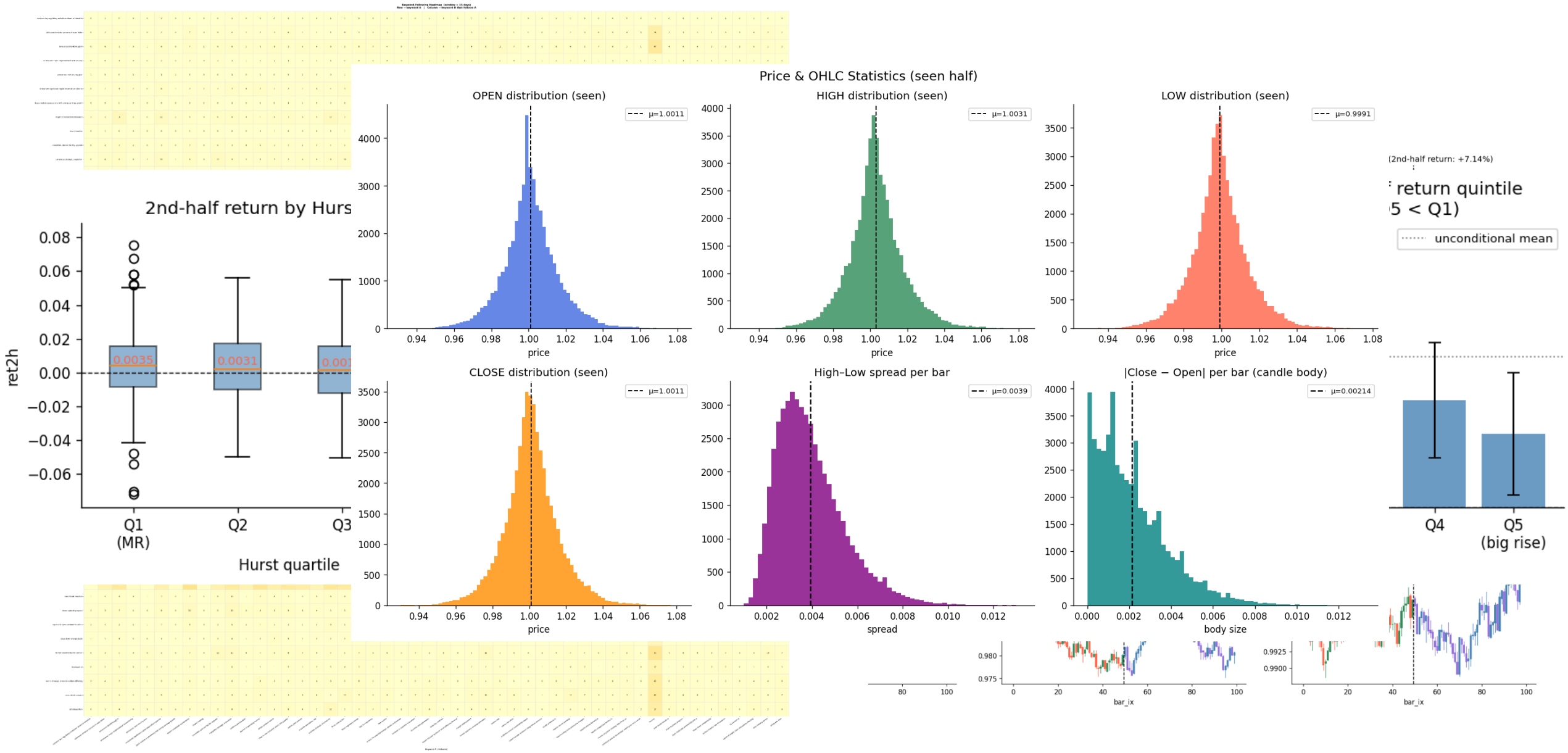

2nd-half return by Hurst quartile
ret2h
0.0035
0.0031
0.0019
0.0056
Q1 (MR)
Q2
Q3
Q4 (trend)
Hurst quartile

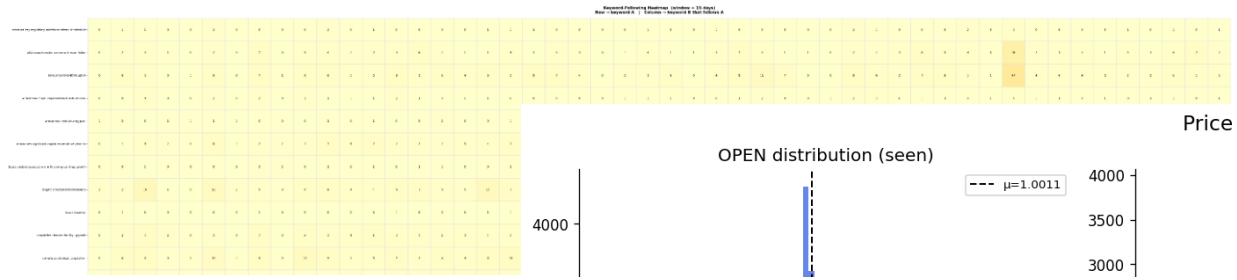

ret_1h vs ret2h (corr=-0.071)
colour = Hurst (red=MR, green=trend)
ret2h
ret_1h
Hurst


Mean ret2h by 1st-half return quintile
(contrarian = Q5 < Q1)
unconditional mean
mean ret2h
Q1 (big drop)
Q2
Q3
Q4
Q5 (big rise)

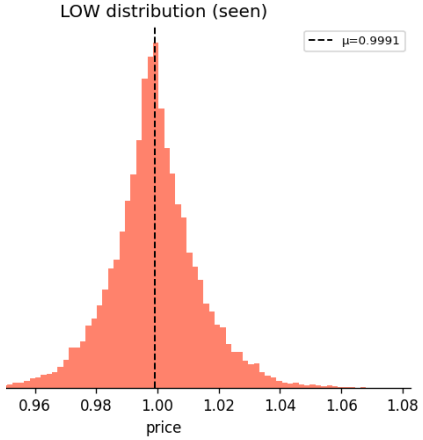



price
bar_ix

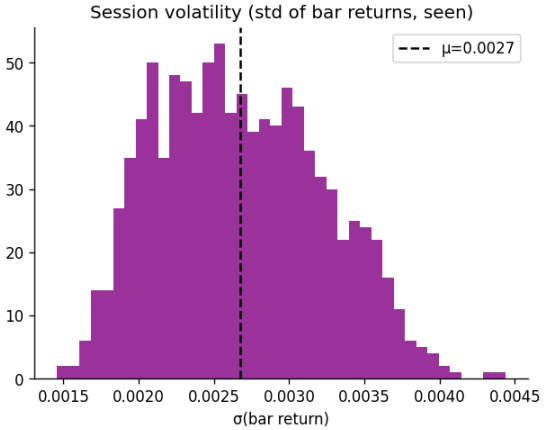

Price & OHLC Statistics (seen half)
OPEN distribution (seen)
μ=1.0011
HIGH distribution (seen)
μ=1.0031
LOW distribution (seen)
μ=0.9991
CLOSE distribution (seen)
μ=1.0011
High–Low spread per bar
μ=0.0039
|Close − Open| per bar (candle body)
μ=0.00214
price
spread
body size
bar_ix
2nd-half return by Hurs
ret2h
0.0035
0.0031
Q1
(MR)
Q2
Q3
Hurst quartile
(2nd-half return: +7.14%)
return quintile
5 < Q1)
unconditional mean
Q4
Q5
(big rise)

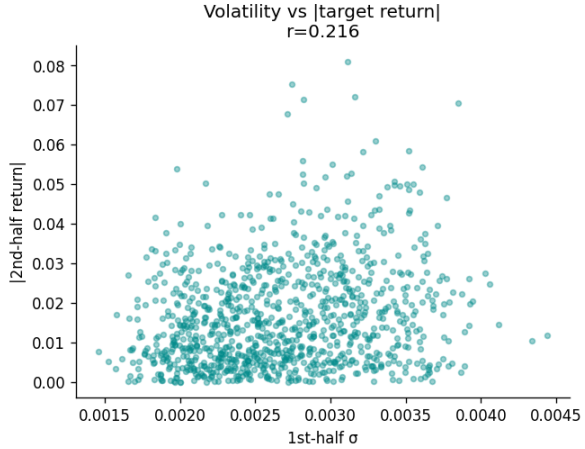


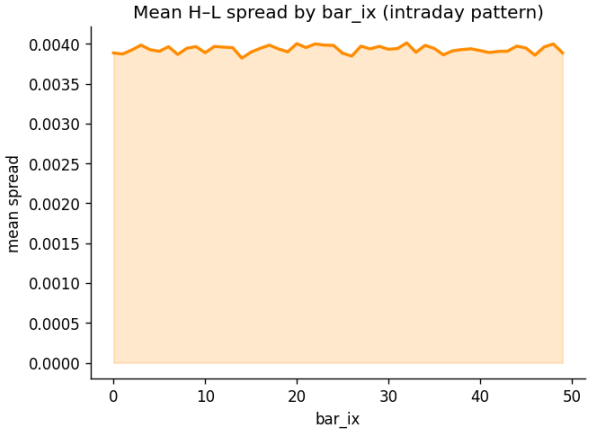
Price & OHLC Statistics (seen half)

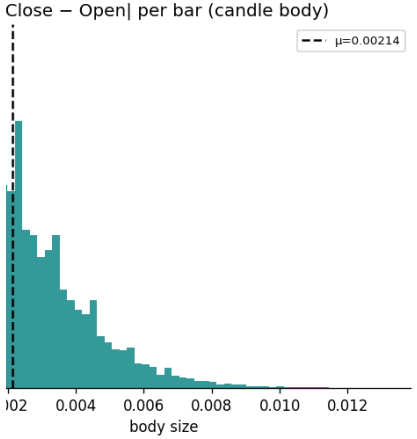


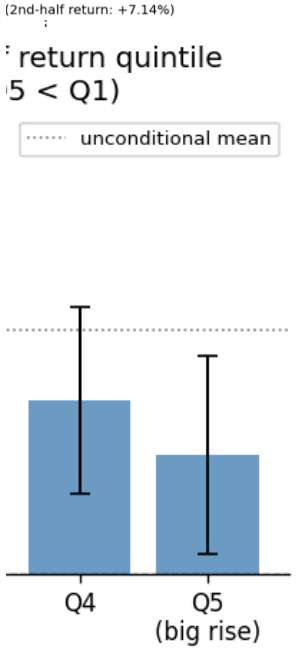
Volatility & Spread Analysis

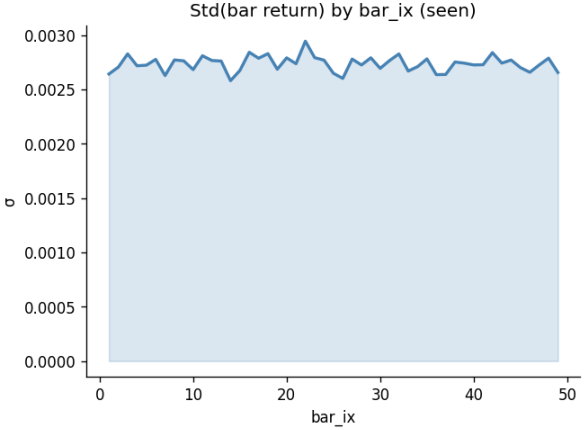


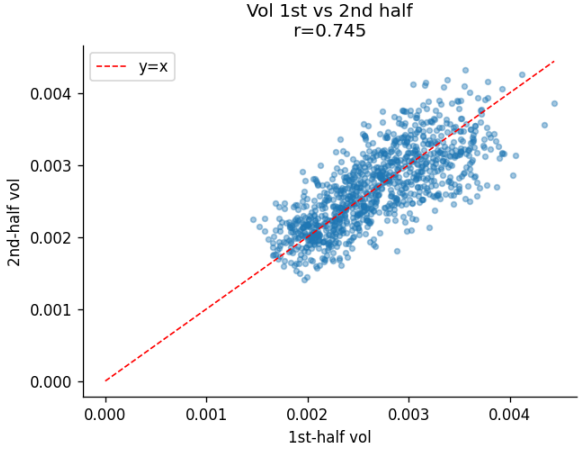


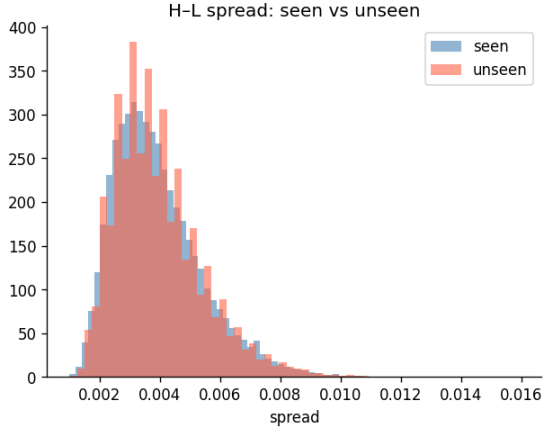


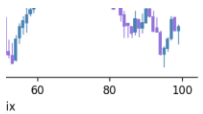


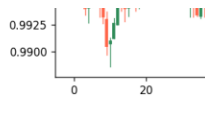


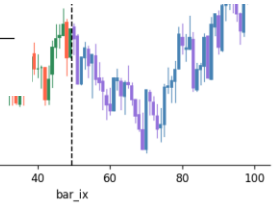


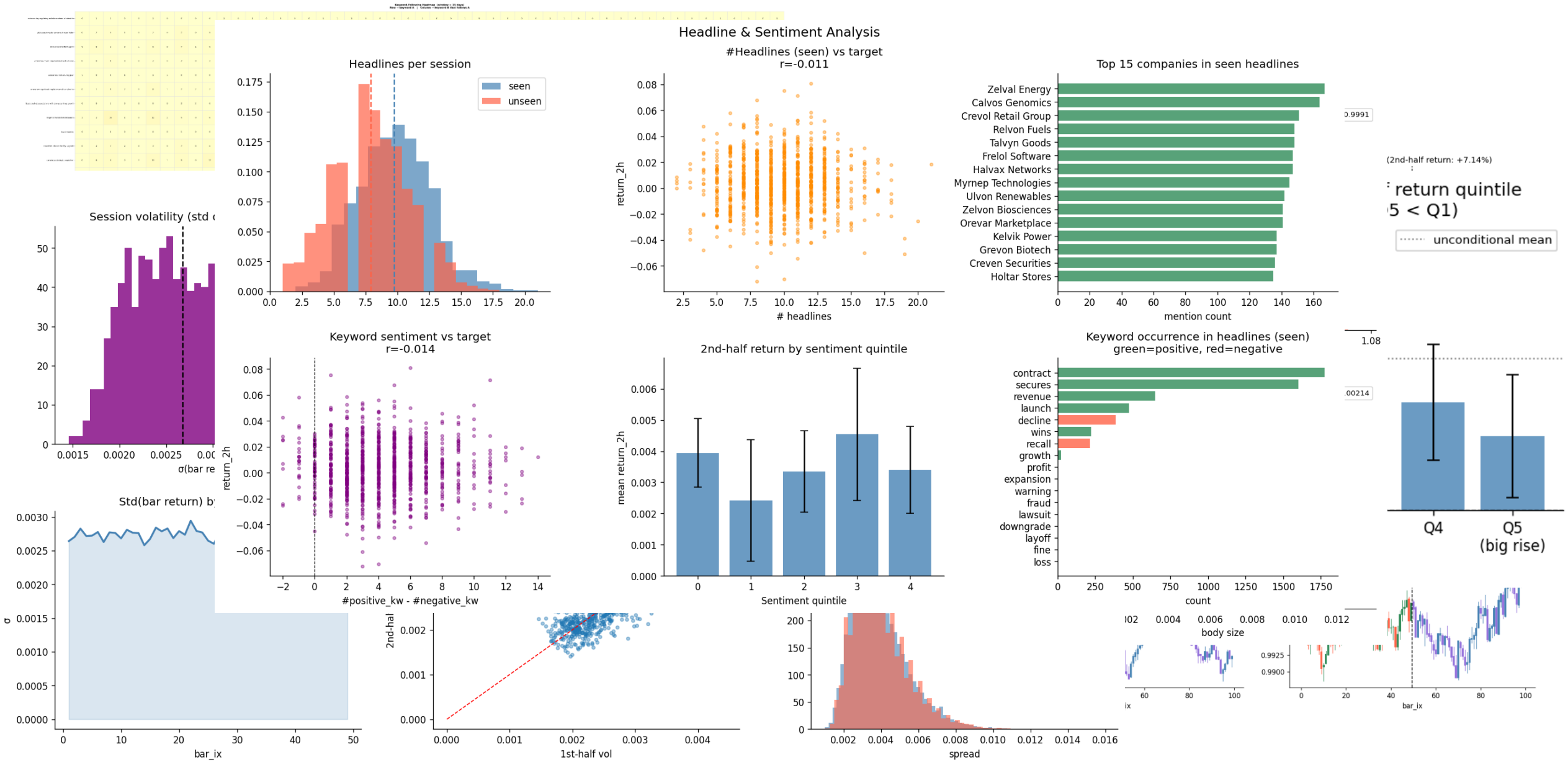




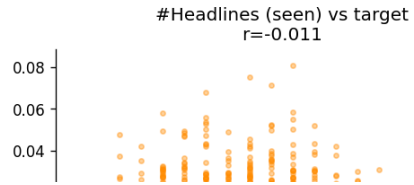


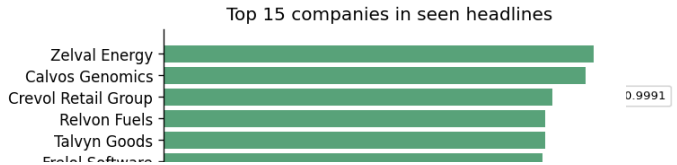




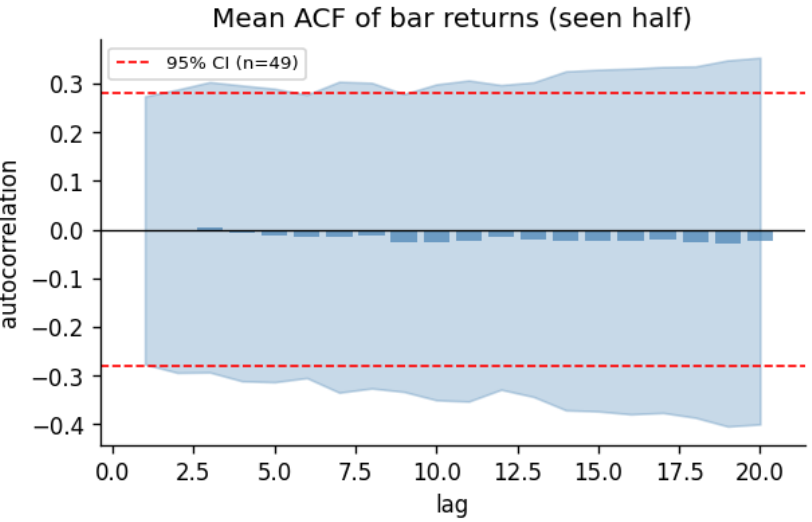

Headline & Sentiment Analysis
Headlines per session
seen
unseen
#Headlines (seen) vs target
r=-0.011
return_2h
headlines
Top 15 companies in seen headlines
Zelval Energy
Calvos Genomics
Crevol Retail Group
Relvon Fuels
Talvyn Goods
Frelol Software
Halvax Networks
Myrnep Technologies
Ulvon Renewables
Zelvon Biosciences
Orevar Marketplace
Kelvik Power
Grevon Biotech
Creven Securities
Holtar Stores
mention count
Session volatility (std
σ(bar re
Keyword sentiment vs target
r=-0.014
#positive_kw - #negative_kw
2nd-half return by sentiment quintile
mean return_2h
Sentiment quintile
Keyword occurrence in headlines (seen)
green=positive, red=negative
contract
secures
revenue
launch
decline
wins
recall
growth
profit
expansion
warning
fraud
lawsuit
downgrade
layoff
fine
loss
count
(2nd-half return: +7.14%)
return quintile
unconditional mean
Q4
Q5
(big rise)
Std(bar return)
σ
bar_ix
2nd-hal
1st-half vol
spread
body size
bar_ix

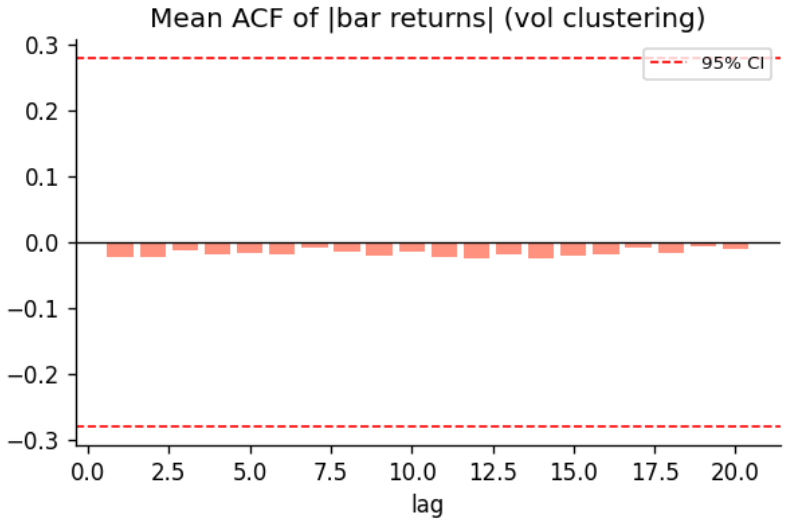
Headline & Sentiment Analysis

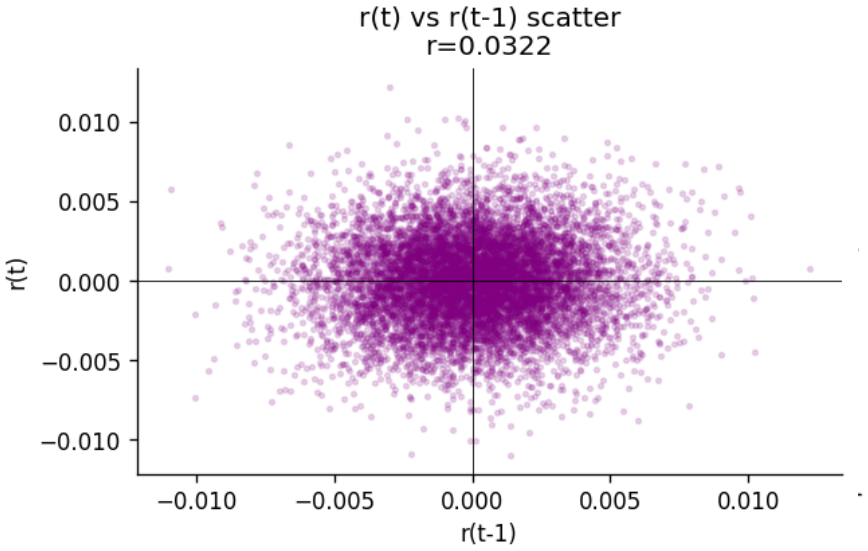


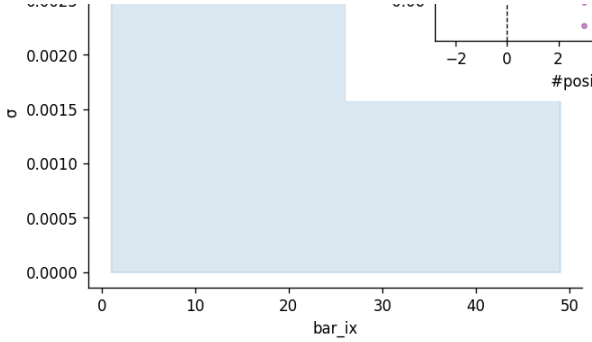


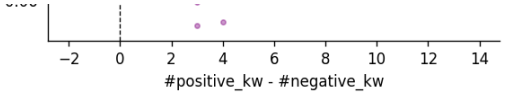
Return Autocorrelation Analysis

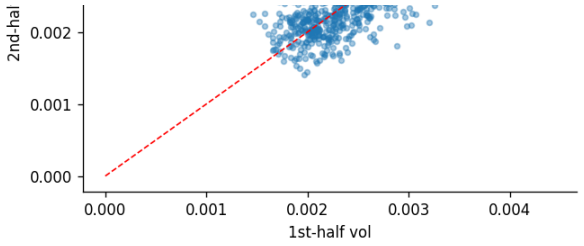


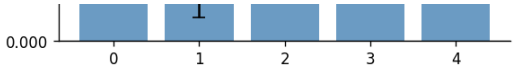


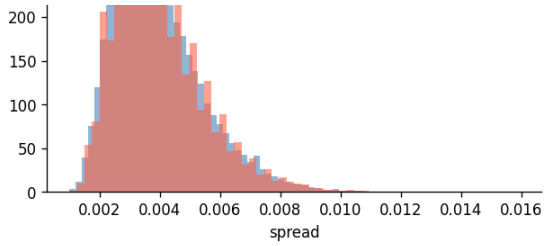


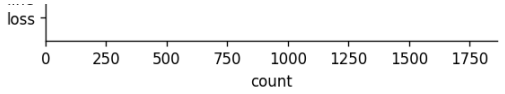


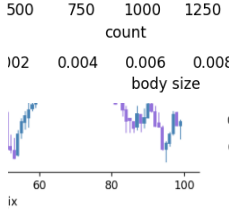


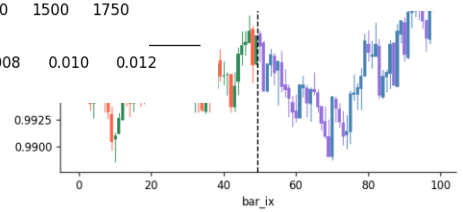


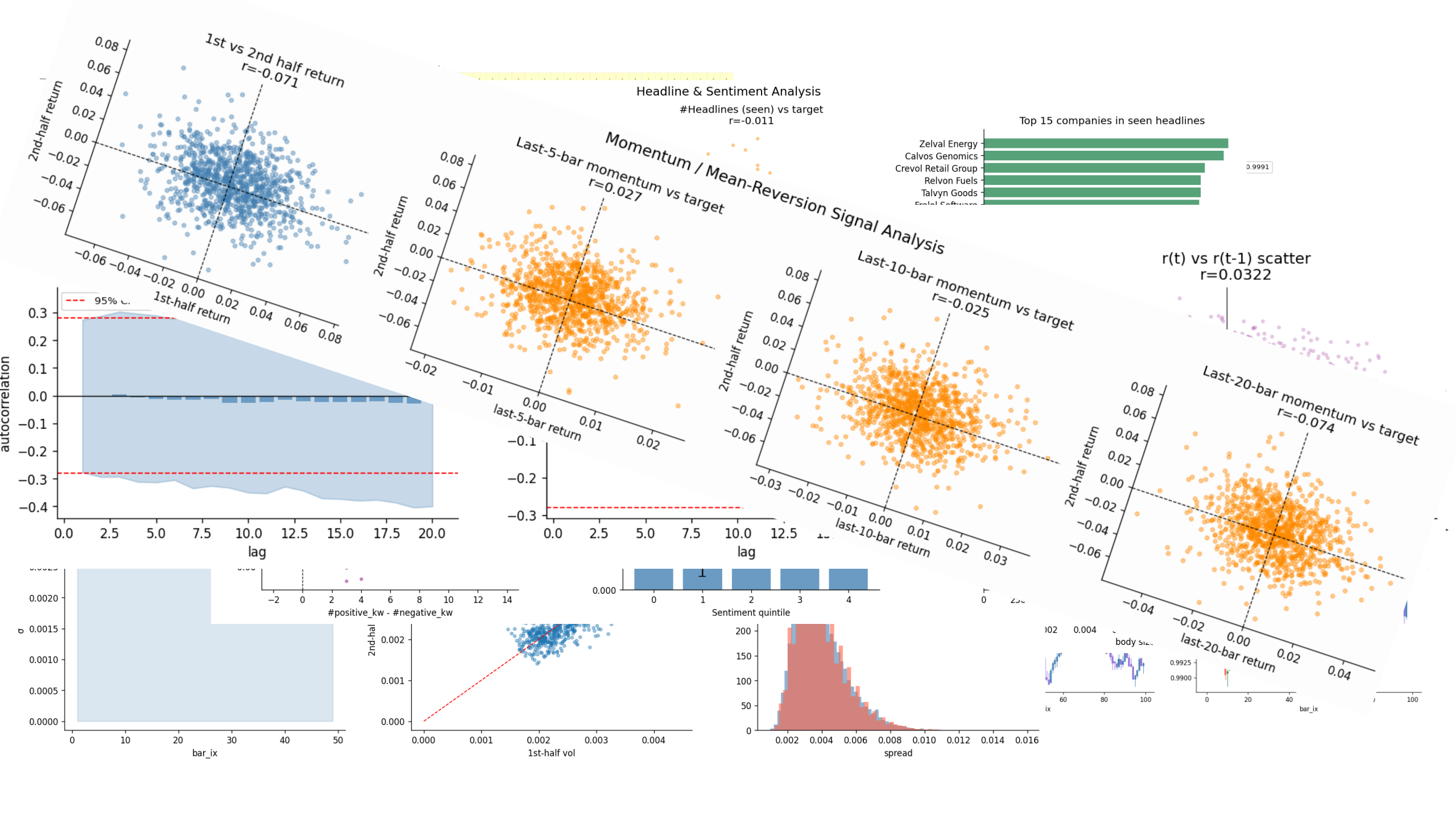


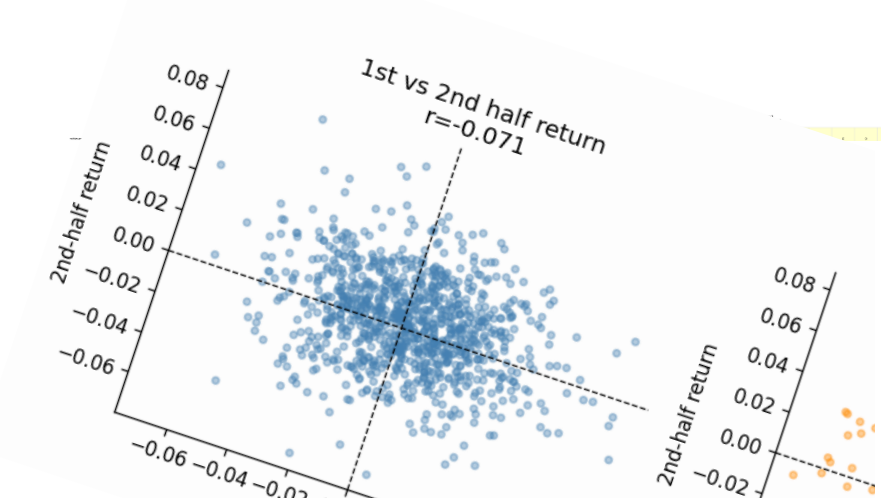


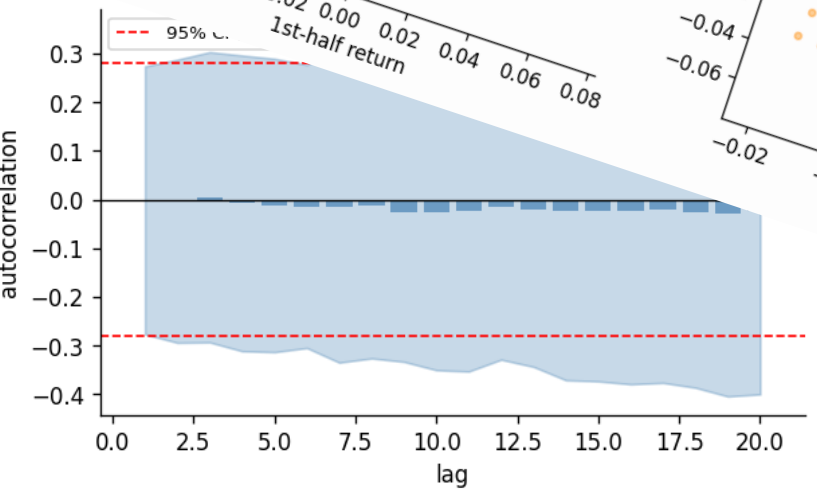


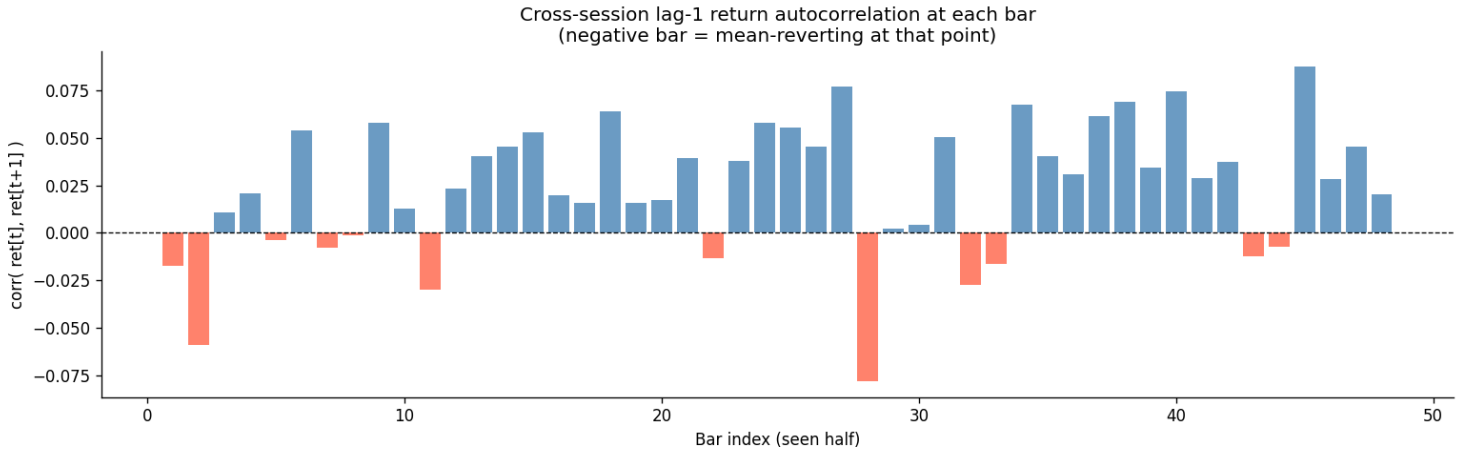




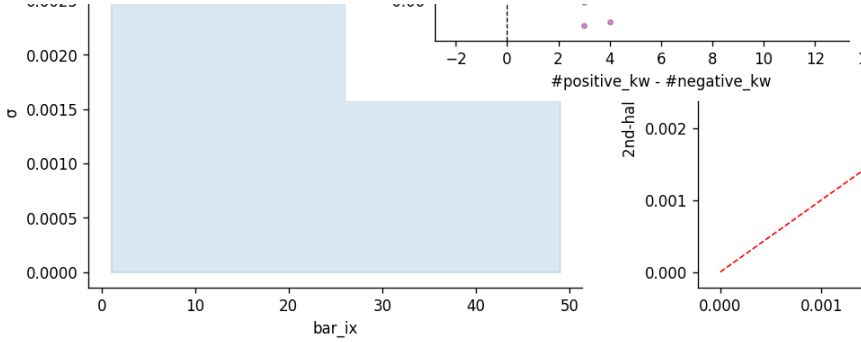

1st vs 2nd half return
r=-0.071
2nd-half return
1st-half return
95% C.
autocorrelation
lag
Momentum / Mean-Reversion Signal Analysis
Last-5-bar momentum vs target
r=0.027
last-5-bar return
Last-10-bar momentum vs target
r=-0.025
last-10-bar return
Last-20-bar momentum vs target
r=-0.074
last-20-bar return
Headline & Sentiment Analysis
#Headlines (seen) vs target
r=-0.011
Top 15 companies in seen headlines
Zelval Energy
Calvos Genomics
Crevol Retail Group
Relvon Fuels
Talvyn Goods
0.9991
r(t) vs r(t-1) scatter
r=0.0322
#positive_kw - #negative_kw
Sentiment quintile
bar_ix
1st-half vol
spread
body size

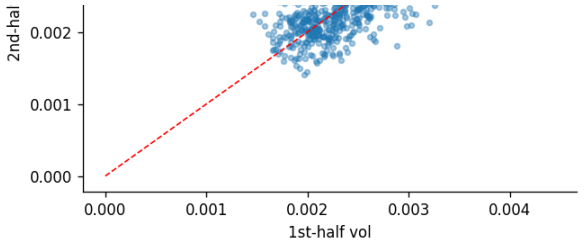

1st vs 2nd half return
r=-0.071
2nd-half return
1st-half return

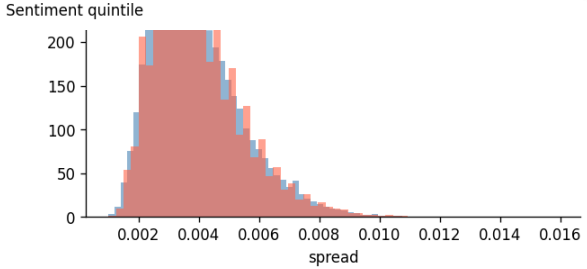

95% C.
autocorrelation
lag

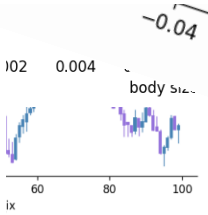

Cross-session lag-1 return autocorrelation at each bar
(negative bar = mean-reverting at that point)
corr(ret[t], ret[t+1])
Bar index (seen half)

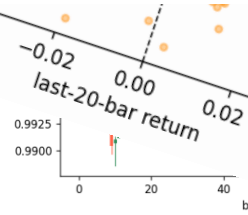
Mean price trajectory across all training sessions
close
open=1.0
Mean close price
Bar index


#positive_kw - #negative_kw
bar_ix

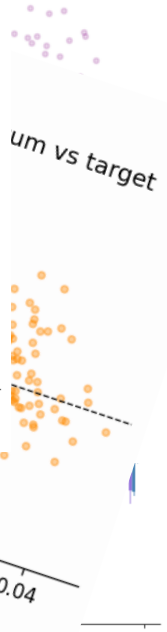

2nd-hal
1st-half vol

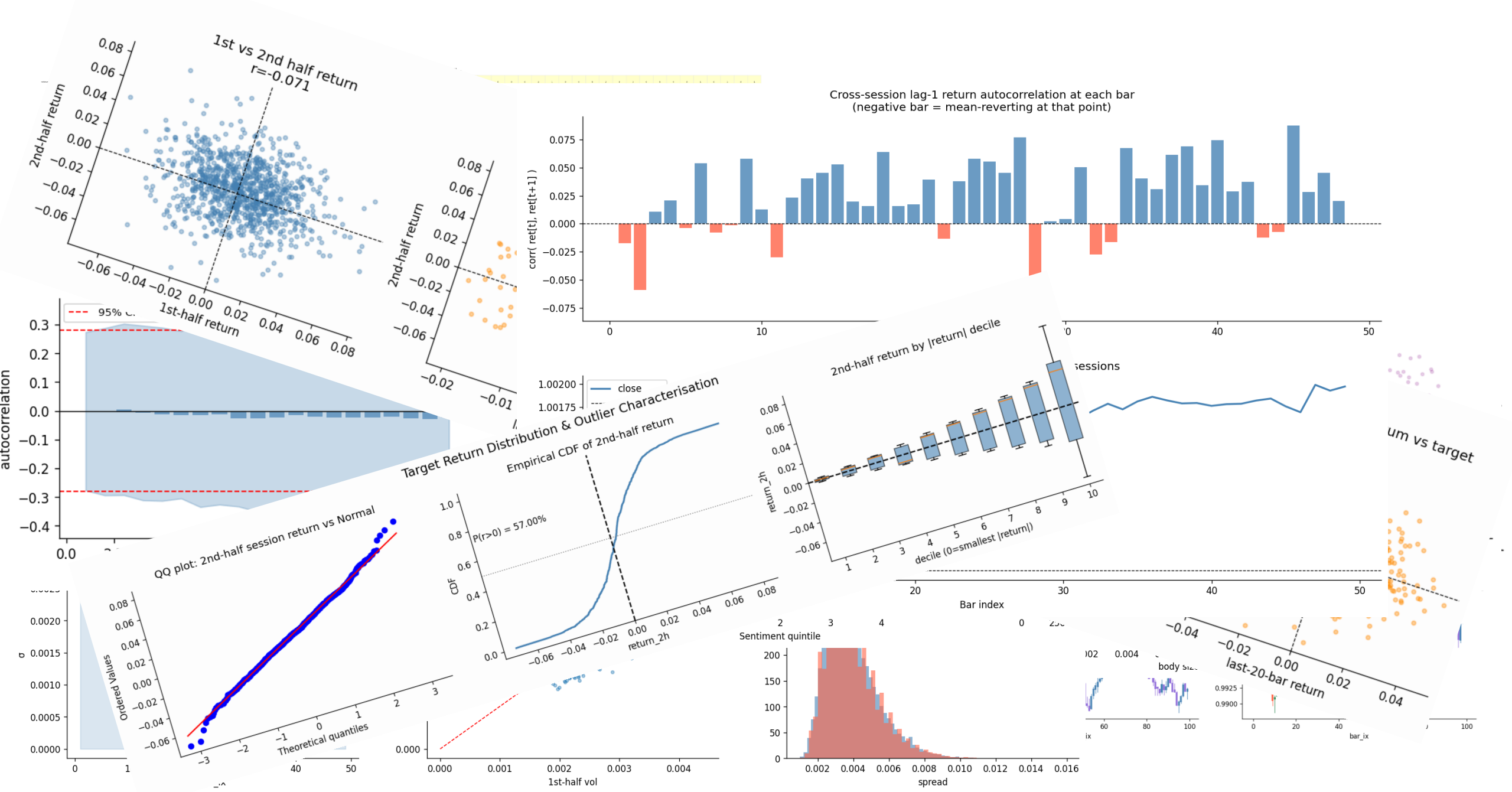

Sentiment quintile
spread

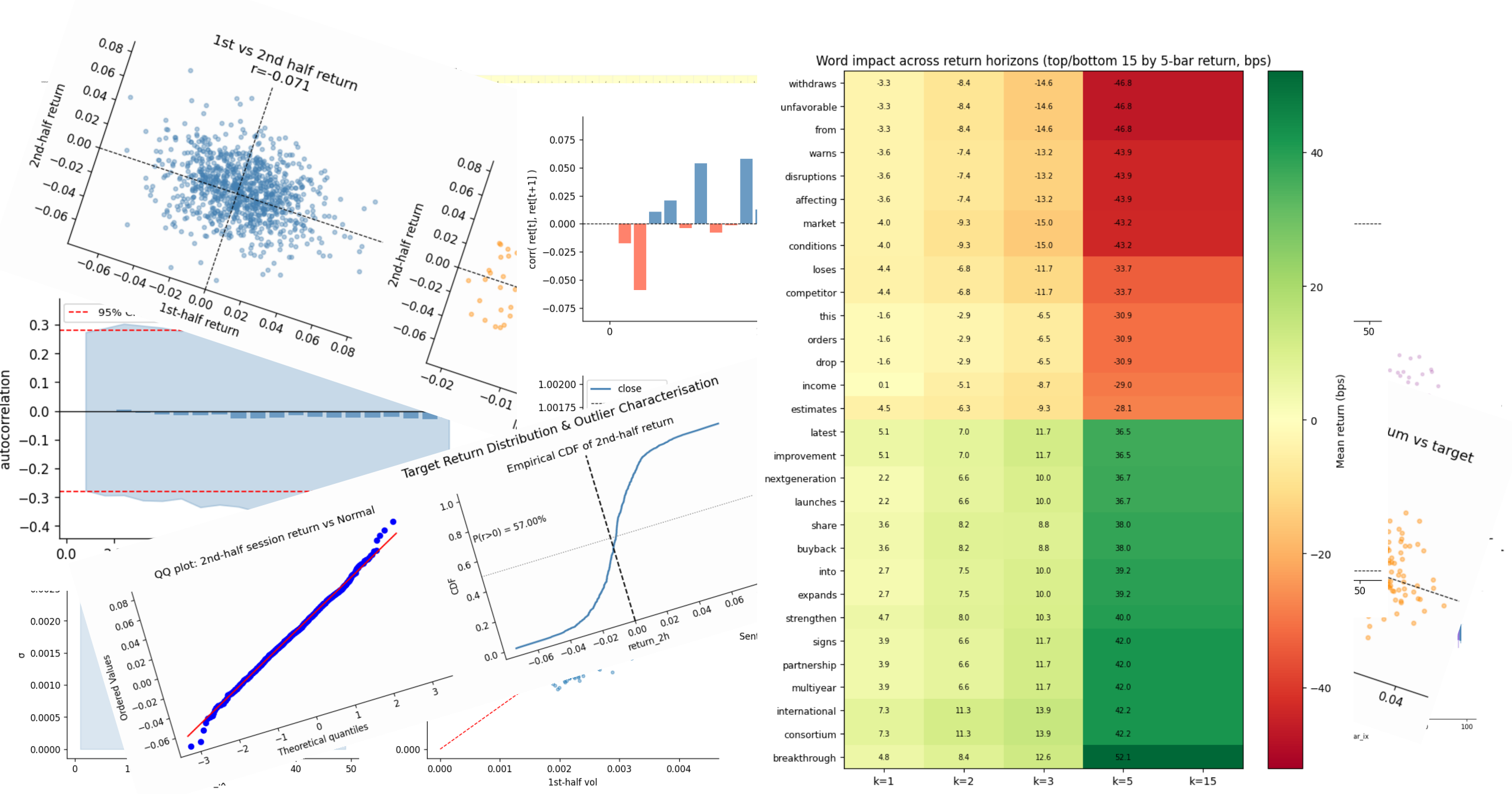

body si
ix


bar_ix

last-20-bar return


um vs target


1st vs 2nd half return
r=-0.071
2nd-half return
1st-half return
Cross-session lag-1 return autocorrelation at each bar
(negative bar = mean-reverting at that point)
corr(ret[t], ret[t+1])
95% C.
autocorrelation
Target Return Distribution & Outlier Characterisation
close
Empirical CDF of 2nd-half return
P(r>0) = 57.00%
CDF
return_2h
2nd-half return by |return| decile
decile (0=smallest |return|)
Bar index
sessions
QQ plot: 2nd-half session return vs Normal
Ordered Values
Theoretical quantiles
1st-half vol
Sentiment quintile
spread
last-20-bar return
bar_ix
body size
um vs target


1st vs 2nd half return
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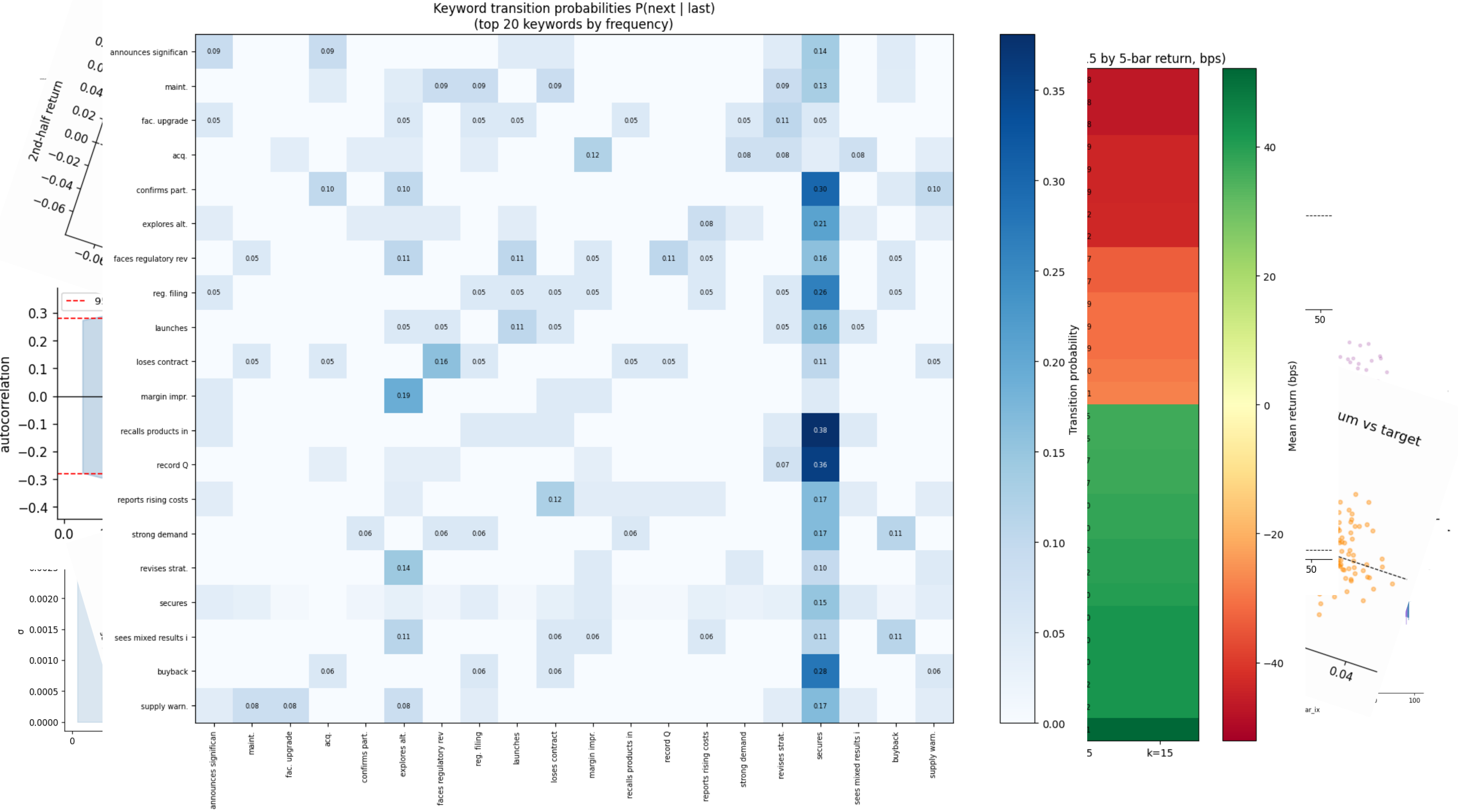
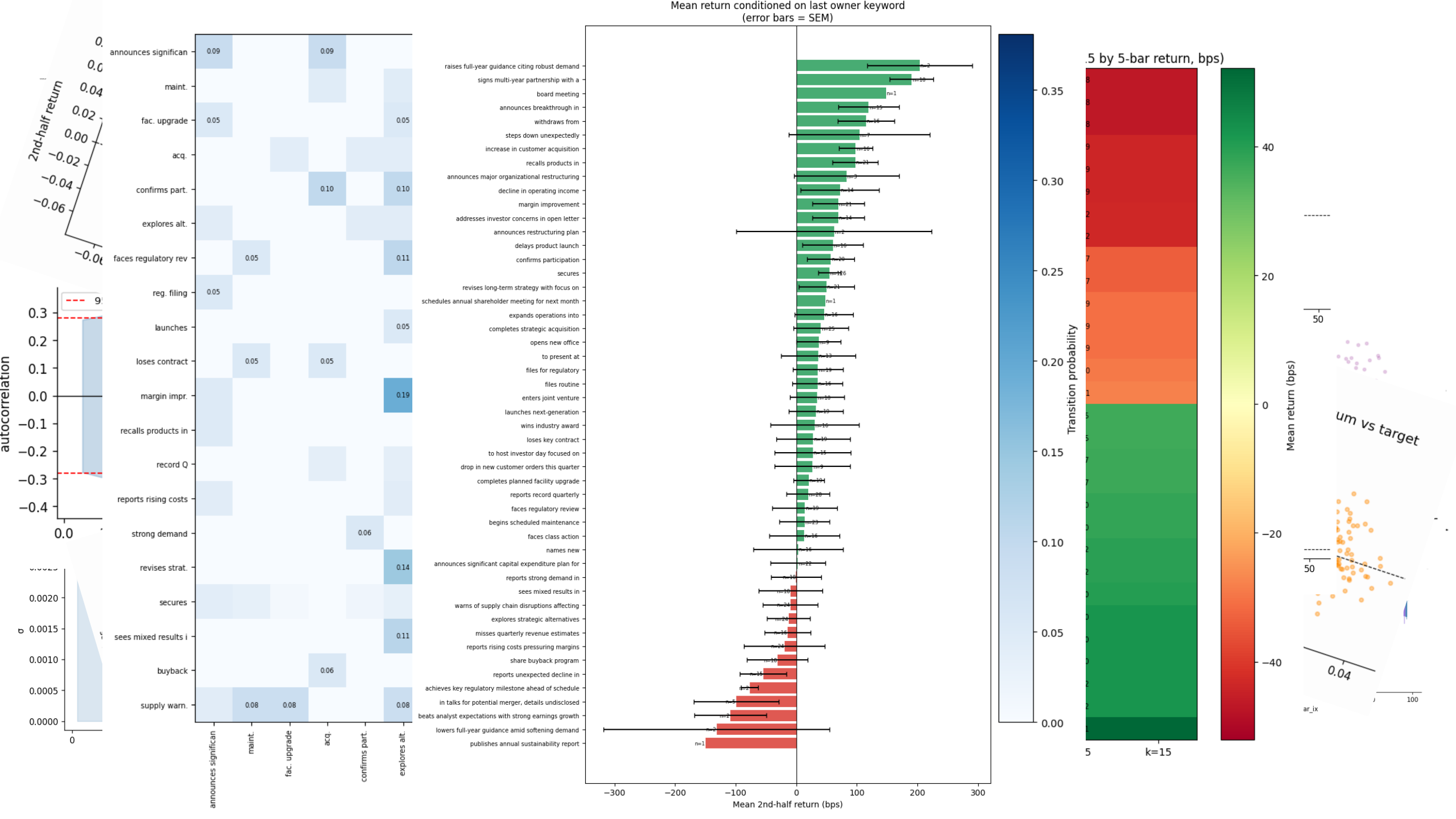
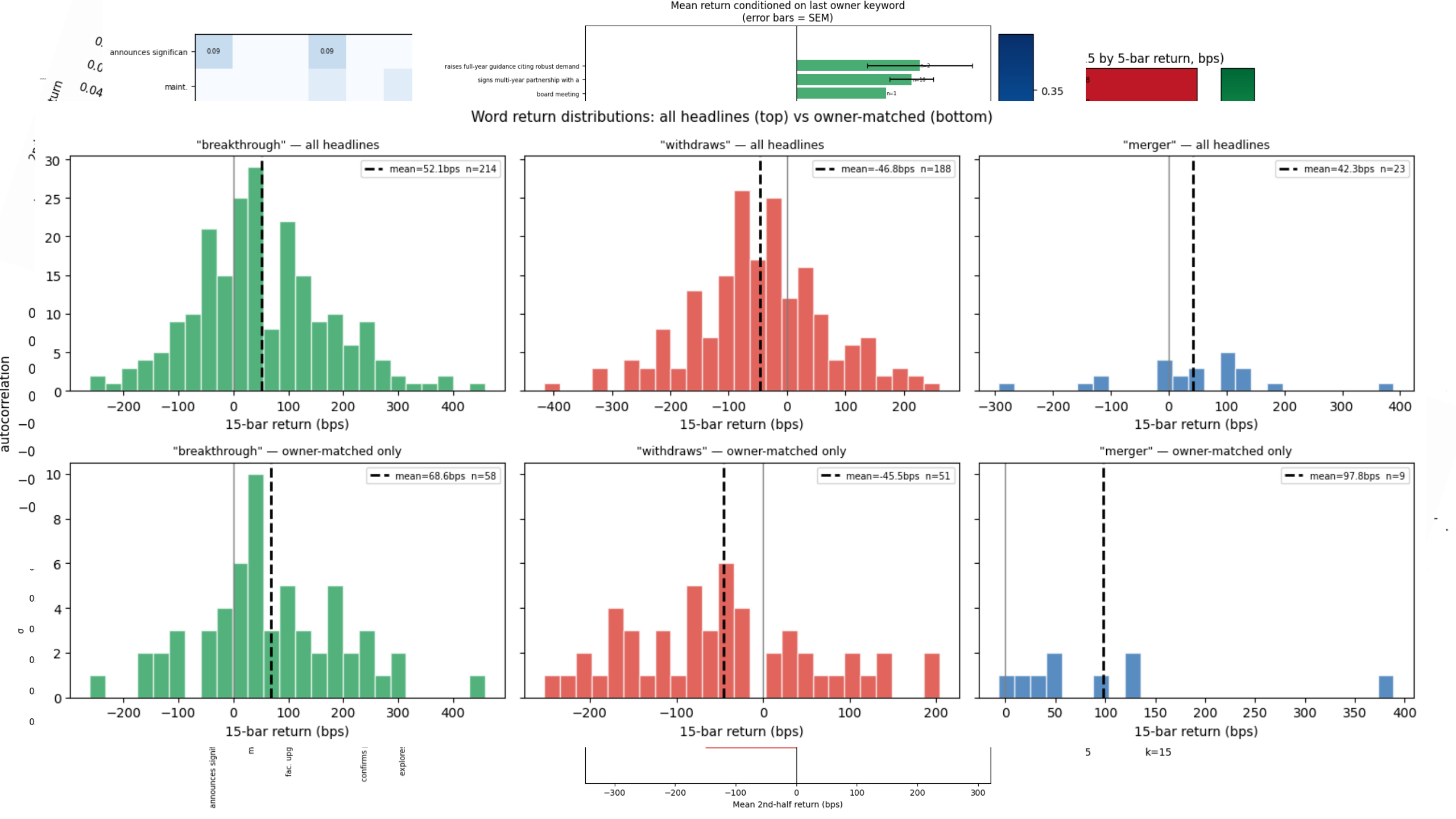
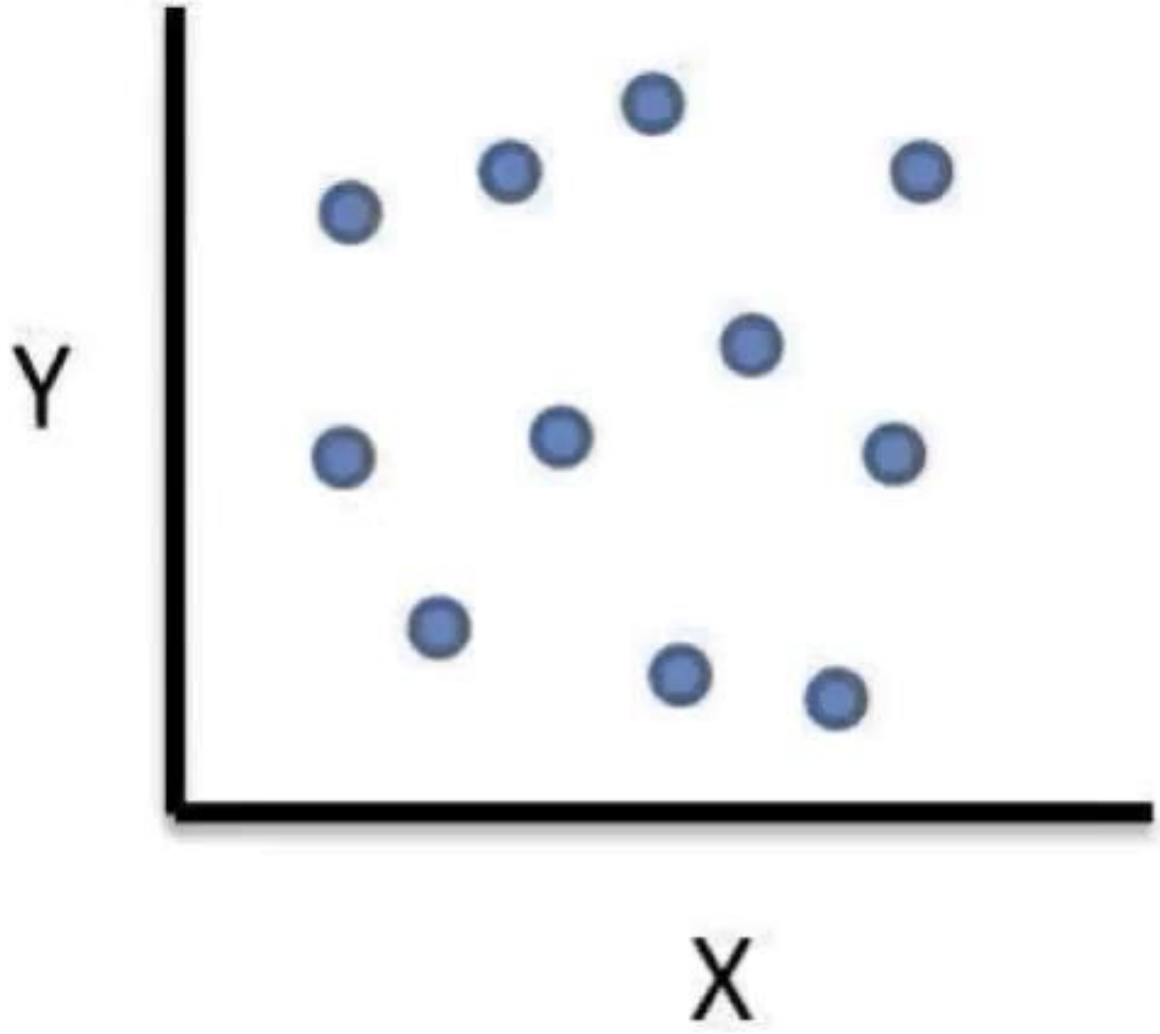
Empirical CDF of 2nd-half return
P(r>0) = 57.00%
CDF
return_2h
QQ plot: 2nd-half session return vs Normal
Ordered Values
Theoretical quantiles
1st-half vol
Word impact across return horizons (top/bottom 15 by 5-bar return, bps)
withdraws -3.3 -8.4 -14.6 -46.8
unfavorable -3.3 -8.4 -14.6 -46.8
from -3.3 -8.4 -14.6 -46.8
warns -3.6 -7.4 -13.2 -43.9
disruptions -3.6 -7.4 -13.2 -43.9
affecting -3.6 -7.4 -13.2 -43.9
market -4.0 -9.3 -15.0 -43.2
conditions -4.0 -9.3 -15.0 -43.2
loses -4.4 -6.8 -11.7 -33.7
competitor -4.4 -6.8 -11.7 -33.7
this -1.6 -2.9 -6.5 -30.9
orders -1.6 -2.9 -6.5 -30.9
drop -1.6 -2.9 -6.5 -30.9
income 0.1 -5.1 -8.7 -29.0
estimates -4.5 -6.3 -9.3 -28.1
latest 5.1 7.0 11.7 36.5
improvement 5.1 7.0 11.7 36.5
nextgeneration 2.2 6.6 10.0 36.7
launches 2.2 6.6 10.0 36.7
share 3.6 8.2 8.8 38.0
buyback 3.6 8.2 8.8 38.0
into 2.7 7.5 10.0 39.2
expands 2.7 7.5 10.0 39.2
strengthen 4.7 8.0 10.3 40.0
signs 3.9 6.6 11.7 42.0
partnership 3.9 6.6 11.7 42.0
multiyear 3.9 6.6 11.7 42.0
international 7.3 11.3 13.9 42.2
consortium 7.3 11.3 13.9 42.2
breakthrough 4.8 8.4 12.6 52.1
k=1
k=2
k=3
k=5
k=15
Mean return (bps)
um vs target


Keyword transition probabilities P(next | last)
(top 20 keywords by frequency)
announces significan
maint.
fac. upgrade
acq.
confirms part.
explores alt.
faces regulatory rev
reg. filing
launches
loses contract
margin impr.
recalls products in
record Q
reports rising costs
strong demand
revises strat.
secures
sees mixed results i
buyback
supply warn.
Transition probability
Mean return (bps)
.5 by 5-bar return, bps)
k=15
um vs target
2nd-half return
autocorrelation


Mean return conditioned on last owner keyword
(error bars = SEM)
Mean 2nd-half return (bps)
raises full-year guidance citing robust demand
signs multi-year partnership with a
board meeting
announces breakthrough in
withdraws from
steps down unexpectedly
increase in customer acquisition
recalls products in
announces major organizational restructuring
decline in operating income
margin improvement
addresses investor concerns in open letter
announces restructuring plan
delays product launch
confirms participation
secures
revises long-term strategy with focus on
schedules annual shareholder meeting for next month
expands operations into
completes strategic acquisition
opens new office
to present at
files for regulatory
files routine
enters joint venture
launches next-generation
wins industry award
loses key contract
to host investor day focused on
drop in new customer orders this quarter
completes planned facility upgrade
reports record quarterly
faces regulatory review
begins scheduled maintenance
faces class action
names new
announces significant capital expenditure plan for
reports strong demand in
sees mixed results in
warns of supply chain disruptions affecting
explores strategic alternatives
misses quarterly revenue estimates
reports rising costs pressuring margins
share buyback program
reports unexpected decline in
achieves key regulatory milestone ahead of schedule
in talks for potential merger, details undisclosed
beats analyst expectations with strong earnings growth
lowers full-year guidance amid softening demand
publishes annual sustainability report
announces significan
maint.
fac. upgrade
acq.
confirms part.
explores alt.
faces regulatory rev
reg. filing
launches
loses contract
margin impr.
recalls products in
record Q
reports rising costs
strong demand
revises strat.
secures
sees mixed results i
buyback
supply warn.
2nd-half return
autocorrelation
Transition probability
Mean return (bps)
k=15
um vs target
ar_ix


Mean return conditioned on last owner keyword
(error bars = SEM)
announces significan
maint.
0.09
0.09
raises full-year guidance citing robust demand
signs multi-year partnership with a
board meeting
0.35
.5 by 5-bar return, bps)
Word return distributions: all headlines (top) vs owner-matched (bottom)
"breakthrough" — all headlines
mean=52.1bps n=214
"withdraws" — all headlines
mean=-46.8bps n=188
"merger" — all headlines
mean=42.3bps n=23
"breakthrough" — owner-matched only
mean=68.6bps n=58
"withdraws" — owner-matched only
mean=-45.5bps n=51
"merger" — owner-matched only
mean=97.8bps n=9
15-bar return (bps)
autocorrelation
Mean 2nd-half return (bps)
k=15

No Correlation



Claude

Claude

Your final answer is your original submission. Submit it now and stop.

Stop here. The pattern is unambiguous.

Stop here. The pattern is unambiguous.

On a more serious note

- Identified all types of headlines
- Sentimental analysis of headlines using FinBERT
- Experimented with finding patterns in causal relationship between them
  - Restructuring announced -> new CEO
  - Going to conference/summit -> contract secured
- Experimented with identifying stock from the session+headlines
- Autocorrelation
- Foundational timeseries model Chronos2

Our best submission

- Ridge regression on features
  - Averages of engineered sentiment features on a session
  - High-low difference means, realized volatitily